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United States Patent Application Publication

20250281036

Kind Code

A1

Publication Date

September 11, 2025

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SYSTEM FOR TREATING EAR DISORDERS

Abstract

Systems and methods can be employed for trans-tympanic membrane access to the middle ear for delivery of a formulation to a targeted site under direct visualization. The systems and methods can also be used to improve accessibility and visualization for various otological surgical procedures, such as, but not limited to, cholesteatoma removal, tympanic membrane repair and ossicular chain repair.

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Family ID: 96948149

Appl. No.: 19/064088

Filed: February 26, 2025

Related U.S. Application Data

us-provisional-application US 63561926 20240306

Publication Classification

Int. Cl.: A61B1/227 (20060101); A61B1/00 (20060101); A61B1/05 (20060101); A61B1/06 (20060101); A61M31/00 (20060101)

U.S. Cl.:

CPC A61B1/227 (20130101); A61B1/00009 (20130101); A61B1/00018 (20130101); A61B1/00043 (20130101); A61B1/00066 (20130101); A61B1/00119 (20130101); A61B1/00128 (20130101); A61B1/05 (20130101); A61B1/0684 (20130101); A61M31/00 (20130101); A61M2202/0007 (20130101); A61M2202/04 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/561,926, filed on Mar. 6, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This document relates to systems and methods for accessing the ear in order to diagnose or treat ear disorders including, but not limited to Meniere's disease and hearing loss. In some examples, the systems and methods include trans-tympanic access through the tympanic membrane for targeted delivery of a therapeutic formulation under direct visualization.

BACKGROUND

[0003] The human ear is subject to a variety of disorders including, but not limited to, hearing loss, balance disorders including vertigo, Meniere's Disease, vestibular neuronitis, vestibular schwannoma, tinnitus, labyrinthitis, otosclerosis, ossicular chain dislocation, cholesteatoma, outer ear infections, middle ear infections, and tympanic membrane perforations, to provide a few examples.

[0004] In one example, SensoriNeural Hearing Loss (SNHL) is due to the absence of, or damage to, hair cells in the cochlea, or to the acoustic nerve. SNHL is typically associated with exposure to loud noise, head trauma, aging, infection, Meniere's Disease, tumors, ototoxicity, genetic diseases like Usher's disease, and the like. Conductive Hearing Loss (CHL) involves the loss of normal mechanical pathways for sound to reach the hair cells in the cochlea, for example due to malformation, accumulation of fluid in the middle ear, disruption of the tympanic membrane, presence of tumors, and/or damage to ossicles. Vestibular disorders include conditions that affect the vestibular system, which is responsible for maintaining balance and providing spatial orientation. The vestibular system is primarily located in the inner ear, meaning that damage to structures within the inner ear can lead to vestibular disorders.

SUMMARY

[0005] This document describes devices, systems, and methods for minimally invasive access to the middle ear for purposes of delivering treatment for inner and/or middle ear disorders. For example, this document describes otologic devices, systems, and methods for minimally invasive, trans-tympanic delivery of a therapeutic formulation to a targeted site within the ear (e.g., to a round window niche of the cochlea) while also providing direct visualization (e.g., using a tip-mounted camera and illumination tool) of such delivery. In particular implementations, the therapeutic formulation can be deposited at the targeted site, and an active agent of the therapeutic formulation can then transfer passively by diffusion across the round window membrane(s), according to a concentration gradient, into the perilymph (within the cochlea). The devices, systems, and methods described herein may be used to treat a variety of disorders of the middle ear and/or inner ear including, but not limited to, hearing loss and balance disorders, including vertigo, Meniere's Disease, vestibular neuronitis, vestibular schwannoma, tinnitus, labyrinthitis, otosclerosis, ossicular chain dislocation, cholesteatoma, middle ear infections, and tympanic membrane perforations, to provide a few examples.

[0006] Some embodiments described herein include an improved instrumentation configured to provide rapid access through a small opening in the tympanic membrane in a manner that provides a clinician with both accurate visualization and enhanced comfort/ease of use during delivery of the treatment. In particular examples described herein, a handheld otology instrument can include a

handle from which first and second shafts extend distally in a side-by-side configuration, with the first shaft having a miniature tip-mounted camera fixed to a distal end of a first shaft and the second shaft having a fluid delivery lumen extending to a distal port of the second shaft. In these examples, the user can readily grasp a handle (e.g., using a pencil grip or other grip that enhances control and accuracy) to through a small radial opening on the tympanic membrane (a myringotomy) using a distal tip of the second (longer) shaft and then advancing both of the side-by-side shafts through the small opening, thereby achieving a micro-incision on the tympanic membrane that is capable of self-healing and substantially the same size as the combined lateral width of the side-by-side shafts. In such examples, the opening formed in the tympanic membrane can be no greater than 2 mm, and preferably about 1 mm to about 1.8 mm. When the shafts are advanced through the opening in the tympanic membrane, the user can then accurately advance the distal port of the second shaft toward the round window niche of the cochlea while the tip-mounted camera at the distal end of the first shaft provides direct visualization of the second shaft.

[0007] While the user retains the handheld otology instrument in a selected position (e.g., with the distal portion of the second shaft proximate to the round window niche and under direct visualization), the user engages an actuator (e.g., using a second hand) to achieve controlled delivery of a therapeutic formulation from the distal port and onto the targeted site. Additionally, in some embodiments, the distal portion of the second shaft can be sized for further insertion beyond the middle ear and into the inner while under direct visualization. For example, the distal portion of the second shaft can, in some embodiments, penetrate the round window membrane such that the distal port of the fluid delivery shaft is positioned within the cochlea to deliver a therapeutic formulation directly into the perilymph.

[0008] In one aspect, this disclosure is directed to a system for delivering a therapeutic formulation at a targeted site in an ear of a patient. The system includes an otologic delivery tool comprising a handle connected to side-by-side trans-tympanic shafts that are fixedly mounted in a stationary position relative to one another and extend distally from the handle. The side-by-side trans-tympanic shafts comprise a visualization shaft and a therapeutic formulation delivery shaft having a distal port positioned distally of the visualization shaft. The system also includes a tip-mounted image capture device fixedly mounted at a distal end of the visualization shaft and being connected with an image data cable extending through a lumen of the visualization shaft proximally toward the handle. The system also includes a fluid input port on the handle to connect with a treatment actuator device to provide fluid communication to a fluid delivery lumen of the therapeutic formulation delivery shaft. The distal port of the therapeutic formulation delivery shaft is configured to deposit the therapeutic formulation at the targeted site in the ear of the patient while the tip-mounted image capture device provides direct visualization of the therapeutic formulation delivery shaft.

[0009] In another aspect, this disclosure is directed to a system for delivering a formulation at a targeted site in an ear of a patient. The system includes a treatment actuator device; and a formulation delivery tool comprising an image capture device and a visualization shaft comprising a visualization shaft distal portion. The image capture device is located at the visualization shaft distal portion. The formulation delivery tool also includes a delivery shaft defining a lumen, the delivery shaft comprising a delivery shaft distal portion extending distally beyond the visualization shaft distal portion. The visualization shaft and the delivery shaft are aligned so that a location of the delivery shaft distal portion is stationary relative to a location of the visualization shaft distal portion. The delivery shaft is configured to deliver, via the lumen from the treatment actuator device, the formulation at the targeted site in the ear of the patient as the image capture device captures the delivery shaft distal portion. The system also includes a flexible tube connecting the treatment actuator device to the lumen of the delivery shaft.

[0010] In yet another aspect, this disclosure is directed to a system for delivering a formulation to a round window niche of a patient. The system includes: a treatment actuator device and a

formulation delivery tool comprising an image capture device and a visualization shaft comprising a visualization shaft distal portion. The image capture device is located at the visualization shaft distal portion. The formulation delivery tool also includes a delivery shaft defining a lumen, the delivery shaft comprising a delivery shaft distal portion extending distally beyond the visualization shaft distal portion. The delivery shaft is configured to deliver, via the lumen from the treatment actuator device, the formulation at the round window niche of the patient as the image capture device captures the delivery shaft distal portion. The system also includes a flexible tube connecting the treatment actuator device to the lumen of the delivery shaft.

[0011] In yet another aspect, this disclosure is directed to a system for delivering a formulation at a targeted site in an ear of a patient. The system includes a formulation delivery tool comprising a visualization shaft and a tip-mounted camera comprising a complementary metal-oxide semiconductor (CMOS) camera sensor, the CMOS camera sensor being mounted on a distal end of the visualization shaft. The formulation delivery tool also includes a delivery shaft configured to deliver the formulation at the targeted site in the ear of the patient as the CMOS camera sensor captures the delivery shaft.

[0012] In yet another aspect, this disclosure is directed to a trans-tympanic ear device for delivering a formulation at a targeted site in an ear of a patient. The trans-tympanic ear device includes a formulation delivery tool comprising an image capture device and a visualization shaft comprising a visualization shaft distal portion. The image capture device is located at the visualization shaft distal portion. The formulation delivery tool also includes a delivery shaft defining a lumen, the delivery shaft comprising a delivery shaft distal portion extending distally beyond the visualization shaft distal portion. The visualization shaft and the delivery shaft are aligned and attached so that a location of the visualization shaft distal portion is stationary relative to a location of the delivery shaft distal portion. The delivery shaft is configured to deliver, via the lumen from a treatment actuator device, the formulation at the targeted site in the ear of the patient as the image capture device captures the delivery shaft distal portion.

[0013] In yet another aspect, this disclosure is directed to an otologic surgery system comprising a handheld otologic instrument including a handle, shafts for trans-tympanic access extending distally from the handle, and a tip-mounted image capture device fixedly mounted at a distal end of one of the shafts.

[0014] In yet another aspect, this disclosure is directed to a trans-tympanic surgery system comprising a handheld otologic instrument including a handle, trans-tympanic shafts that are fixedly mounted in a stationary position relative to one another and extend distally from the handle. The trans-tympanic shafts comprise a visualization shaft and a therapeutic formulation delivery shaft having a distal port positioned distally of the visualization shaft. The trans-tympanic surgery system also includes a therapeutic formulation contained with a reservoir that is connectable to the handle to provide fluid communication to a fluid delivery lumen of the therapeutic formulation delivery shaft, wherein the distal port of the therapeutic formulation delivery shaft is configured to deposit the therapeutic formulation at a targeted site in an ear of a patient.

[0015] In yet another aspect, this disclosure is directed to an otologic surgery system comprising a disposable handheld otologic instrument comprising first and second trans-tympanic shafts that are fixedly mounted in a stationary position relative to one another and have a combined lateral width of less than 2 mm. The otologic surgery system also includes a reusable display controller configured to detach from a connection with the disposable handheld otologic instrument and to connect with a second disposable handheld otologic instrument.

[0016] In yet another aspect, this disclosure is directed to a method of treating an ear, comprising advancing trans-tympanic shafts of a handheld otologic instrument through a tympanic membrane while the trans-tympanic shafts are fixedly mounted in a stationary position relative to one another and extend distally from a distal end of a handle of the handheld otologic instrument. The method also comprises using a tip-mounted camera at a distal end of a first shaft of the trans-tympanic

shafts to provide imaging of a distal tip of a second shaft of the trans-tympanic shafts while a delivered amount of a therapeutic formation is deposited out of the distal tip of the of the second shaft and at a targeted site within the ear.

[0017] In yet another aspect, this disclosure is directed to a method of using an otologic surgery system, comprising disconnecting a disposable handheld otologic instrument from a reusable video controller after the video controller is used to display image data of a trans-tympanic shaft of the disposable handheld otologic instrument depositing a delivered amount of a therapeutic formation at a targeted site within an ear. The method also comprises discarding the disposable handheld otologic instrument after a single use while an undelivered amount of the therapeutic formation remains in an interior lumen of the trans-tympanic shaft of the disposable handheld otologic instrument.

[0018] In yet another aspect, this disclosure is directed to a system for delivering a therapeutic formulation at a targeted site in an ear of a patient. The system includes an otologic delivery tool comprising a handle having a distal end and a proximal end and a visualization shaft fixedly connected to the handle and extending distally from the distal end of the handle to a tip-mounted image capture device attached to a distal wall of the visualization shaft and oriented in a distal facing direction, the tip-mounted image capture device being connected with an image data cable extending through a lumen of the visualization shaft proximally toward the handle. The otologic delivery tool also includes a therapeutic formulation delivery shaft configured to extend parallel the visualization shaft and distally from the distal end of the handle toward a distal fluid port along a distal tip of the therapeutic formulation delivery shaft. The distal tip of the therapeutic fluid delivery shaft is positionable distally of the tip-mounted image capture device of the visualization shaft such that, during delivery of a therapeutic formulation from the distal port of the therapeutic formulation delivery shaft, the tip-mounted image capture device provides image data of the distal tip of the therapeutic formulation delivery shaft via the image data cable extending through the lumen of the visualization shaft proximally toward the handle.

[0019] Some or all of the embodiments described herein may provide one or more of the following advantages. First, the systems and methods described herein can include specialized techniques and instruments that achieve rapid and miniature trans-tympanic access (with a small opening formed in the tympanic membrane) to precisely deliver a therapeutic formulation to a targeted site in the middle or inner ear. The systems and methods can be used for minimally invasive ear surgery with a mere local anesthetic (without general anesthesia) to treat a variety of hearing loss conditions or ear disorders, including treatments that precisely place a sustained-delivery therapeutic formulation on or across the round window membrane or to other parts of the cochlea.

[0020] Second, the systems and methods described herein can advantageously provide direct visualization during delivery of the formulation to the targeted site in the middle or inner ear. Optionally, the direct visualization can be provided via a miniature camera-on-a-chip (e.g., a complementary metal-oxide semiconductor (CMOS)) image sensor having width of less than 1 mm, and preferably less than 0.6 mm, mounted at a distal end of an instrument shaft, thereby achieving a smaller overall instrument size (e.g., for the component(s) inserted through the tympanic membrane). The use of such direct visualization advantageously allows visual confirmation of the proper placement of the formulations with a high level of accuracy. The direct visualization also provides additional benefits such as the ability to ascertain visually whether there are any obstructions that could inhibit the proper delivery of the formulations. For example, in some cases the round window is covered by a pseudo membrane that can be altered or moved to allow improved access to the round window niche. By using the improved instrumentation described herein, the presence of the pseudo membrane can be visually verified, and thereafter physically altered, moved, or navigated around, so that improved and direct access to the round window membrane can be established for more precise placement of the formulation. In addition, after the formulation has been administered, direct visualization can be used to verify that the

formulation is retained in the desired position and manner.

[0021] Third, the systems and methods described herein allow direct access to the middle ear cavity through the tympanic membrane in a suture-less, low impact manner. In some implementations, such direct access through the tympanic membrane using a set of side-by-side shafts fixed to an instrument handle can be safer, less invasive, and achieved with no sealing or patching of the tympanic membrane. For example, due to the small size of the side-by-side shafts in some examples described herein, the tympanic membrane can heal naturally after withdrawal of the shafts.

[0022] Fourth, the systems and methods described herein can advantageously provide improved comfort and ease-of-use for a clinician both during the insertion of an instrument into the ear and during the actuation for delivery of the therapeutic formulation to the targeted site in the middle or inner ear. In particular examples, the systems and methods can include a handheld instrument including a handle configured to be gripped in a manner that provides enhanced dexterity and control for the user and that provides an improved size relative to the distal extending shaft(s) that extend from the handle. Optionally, the user can controllably retain the handheld instrument in a precise and preferred position using a first hand (e.g., with a pencil grip upon the handle) while separately acting upon an actuator (e.g., using a second hand) to achieve controlled delivery of the therapeutic formulation to the targeted site in the middle or inner ear.

[0023] Fifth, the systems and methods for treating hearing loss and other ear disorders as described herein facilitate treatments in a minimally invasive fashion. Such minimally invasive techniques allow a clinician to deliver the formulation without an operating room the associated personnel and resources, can tend to reduce recovery times, reduce a risk of complications, reduce patient discomfort, and reduce treatment costs. Moreover, the methods described herein can be performed using a local anesthetic rather than requiring general anesthesia. Accordingly, the treatment cost, patient risks, and recovery times are further advantageously reduced.

[0024] Sixth, the systems and methods described herein can include a disposable delivery tool that does not require sterilization for reuse. For example, in particular embodiments, the delivery tool can be a one-time-use instrument that is readily discarded while other components (such as the display controller or other structures connected to the delivery tool) are reused with subsequent delivery tools in subsequent surgical procedures. In such examples, the fluid delivery shaft of the delivery tool can be attached with the handle but need not evacuate all of the therapeutic formulation from the interior of the shaft into the patient's ear. Rather, the user may deposit only the amount that is sufficient within the user's ear, and an undelivered portion can remain in the interior of the fluid delivery shaft (connected to the handle) for safe and sanitary discarding of all such elements. Such a solution can be particularly effective, for example, where the therapeutic formulation is a self-gelling composition, and the undelivered amount remaining in the interior of the fluid delivery shaft (connected to the handle) may form a gel or harden. According to some embodiments detailed below, the entire delivery tool can be promptly discarded (e.g., including the handle, both shafts, the tip-mounted camera device, and the undelivered portion of the therapeutic formulation within the fluid delivery shaft), thereby advantageously eliminating the need to clean or remove such gel or hardened material from the interior of the fluid delivery shaft. Moreover, such disposable embodiments described herein can improve patient safety because a one-use tool can be thoroughly sterilized, packaged, and subsequently discarded after a single use without a need to sterilize the tool for another use. Using a disposable delivery tool can also improve ease of use by allowing the clinician to dispose of the tool without needing to replace any parts for another use.

[0025] Seventh, the systems described herein can also be used for diagnostic purposes. Such uses can help in procedure planning, change site of care, and potentially improve patient outcomes.

[0026] The details of one or more embodiments of the invention are set forth in the accompanying

drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a perspective view of an ear treatment system, in accordance with some embodiments.

[0028] FIG. 2 is another perspective view of the ear treatment system of FIG. 1, in accordance with an example use with a patient.

[0029] FIG. 3 is a perspective view of a distal portion of a delivery tool of the ear treatment system of FIG. 1, in accordance with some embodiments configured for trans-tympanic access toward a targeted site in the middle or inner ear.

[0030] FIG. 4 shows a first side view of the delivery tool of the ear treatment system of FIG. 1.

[0031] FIG. 5 shows a top side view of the delivery tool of FIG. 4.

[0032] FIG. 6 shows a cross-sectional view of the delivery tool of FIG. 4.

[0033] FIG. 7 shows a perspective view of the delivery tool of FIG. 4.

[0034] FIG. 8 shows a perspective view of a portion of the delivery tool of FIG. 7.

[0035] FIG. 9 shows another perspective view of the delivery tool of FIG. 7.

[0036] FIG. 10 shows an example cross-section of the delivery tool of FIG. 7 including the first shaft and the second shaft.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0037] Referring now to FIG. 1, some embodiments of an ear treatment system **100** can be used to access an ear of a patient **10** through a small opening in the tympanic membrane. Ear treatment system **100** can include a delivery tool **110**, a treatment actuator device **130** (e.g., a fluid delivery syringe in the depicted embodiment), and a display device **150** for providing onscreen visualization of the delivery tool **110**. A distal portion **111** of the delivery tool **110** can extend through the small opening in the tympanic membrane. As described in detail below, the delivery tool **110** can be implemented as a handheld otologic instrument having a handle configured to be retained by a user **1** (external to the ear) and one or more shafts that extend distally from the handle for insertion in a trans-tympanic path (e.g., through an opening formed in the tympanic membrane) to deliver a therapeutic formulation to a targeted site of the patient **10**. As described herein, in some cases the therapeutic formulation is a liquid or gel, and the targeted site is a round window niche of a cochlea **50**. It should be understood from the description here that, in some embodiments, the targeted site can be another location in the middle or within the inner ear. For example, delivery tool **110** can include a fluid delivery shaft that is sized to penetrate the round window membrane and deliver a therapeutic formulation directly into the cochlea, such as into the perilymph of the inner ear. Also as described in more detail below, the delivery tool **110** can be equipped with tip-mounted camera **112** fixed to a distal end of a first shaft **114**, which can advantageously provide direct visualization of the delivery of the therapeutic formulation to the targeted site via a distal port **115** located at a distal end of a second shaft **116**.

[0038] For example, in use during some implementations, the clinician can readily grasp a handle of the delivery tool **110** (e.g., using a pencil grip or other grip that enhances control and accuracy) to simultaneously insert side-by-side shafts of the delivery tool **110** through a small opening formed in the tympanic membrane, such a surgical opening that is no greater than 2 millimeters (mm) and preferably about 1 mm to about 2 mm. Under direct visualization provided by the tip-mounted camera **112** of the delivery tool that is in communication with the display device **150** (via image data cable **142** and display controller **140** in the depicted embodiment), the clinician **1** can grasp the handle of the delivery tool **110** to accurately advance a distal port of the second (longer)

shaft **116** toward the round window niche of the cochlea while the tip-mounted camera **112** at a distal end of first (shorter) shaft **114** provides direct visualization of the second shaft **116**. From there, while the clinician **1** uses a first hand to retain the delivery tool **110** in its operative position (with the distal port **115** of the second shaft **116** proximate to the round window niche and under direct visualization), the clinician **1** can engage the treatment actuator device **130** using a second hand to achieve controlled delivery of the therapeutic formulation from the distal port.

[0039] Still referring to FIG. **1**, the ear treatment system **100** described herein can be in particular methods to treat and/or prevent a variety of ear conditions, including but not limited to hearing loss (such as hidden hearing loss, noise-induced hearing loss, age-related hearing loss, drug-induced hearing loss (e.g., chemotherapy-induced hearing loss or aminoglycoside-induced hearing loss), sudden sensorineural hearing loss (SNHL), autoimmune inner ear disease, and the like) or particular disorders of the middle ear and/or inner ear (such as tinnitus, balance disorders including vertigo, Meniere's Disease, vestibular neuronitis, vestibular schwannoma, labyrinthitis, otosclerosis, ossicular chain dislocation, cholesteatoma, and middle ear infections). Additionally, as described below, the ear treatment system **100** can be used in such a manner to treat a variety of ear conditions while also providing enhanced comfort to the clinician **1** and real-time direct visualization during delivery of the treatment in the middle or inner ear.

[0040] Referring now to FIGS. **1-3**, the delivery tool **110** in the ear treatment system **100** can include a combination of structures that are sized in a manner that is particularly suited for the anatomy of the ear and for minimally invasive approach during insertion through the tympanic membrane **30**. In the depicted embodiment, the delivery tool **110** is manipulated by the clinician **1** as a handheld instrument having first and second shafts **114**, **116** that are contemporaneously advanceable through an external ear canal **38**, through the tympanic membrane **30**, and toward the cochlea **50** of the patient **10**. As depicted in FIGS. **2-3**, the patient **10** can be positioned so as to orient the external ear canal **38** upwardly toward the clinician (and preferably so that the round window niche **52** of the cochlea **50** can be positioned below a distal port of the delivery tool **110** such that the therapeutic fluid is deposited and remains at the round window niche **52**).

[0041] The trans-tympanic access in the depicted embodiment is minimally invasive and can avoid large, irreparable openings or tears in the tympanic membrane **30**. The tympanic membrane **30**, sometimes referred to as the eardrum, represents a thin membrane that separates the external ear canal **38** from middle ear **40**. Tympanic membrane **30** plays a role in transmission of sound. When sound waves enter the external ear canal **38**, the sound waves cause tympanic membrane **30** to vibrate, thus transmitting the vibrations to middle ear **40**. The vibration frequency of tympanic membrane **30** may correspond to a frequency of the sound waves entering the external ear canal **38**. Middle ear **40** is a space between tympanic membrane **30** and the inner ear. As described in more detail below, a distal portion **111** of the delivery tool **110** that is inserted through the tympanic membrane **30** (e.g., the side-by-side shafts **114**, **116** depicted in FIG. **1** and FIG. **3**) can have a maximum lateral width that is reduced to achieve a minimally invasive access that promotes self-healing of the tympanic membrane. For example, a needle puncture is generally less invasive than a larger surgical incision. Delivery tool **110** can be sized to perform a procedure deliver a formulation to a targeted site in an ear of the patient **10** in a way that is more like a needle puncture than a surgical incision, thus limiting an invasiveness of the procedure.

[0042] The middle ear and inner ear regions are positioned internally to the tympanic membrane **30** (opposite from the external ear canal **38**). The middle ear **40** includes bones known as the ossicles such as the malleus, incus, and stapes. The ossicles amplify sound vibrations of tympanic membrane **30** and transmit these vibrations to the inner ear. The middle ear **40** is an air-filled space and is not necessarily filled with fluid. The inner ear, sometimes referred to as the labyrinth, represents a complex structure located within a temporal bone of the skull. The inner ear includes cochlea **50**. Cochlea **50** contains sensory organs for detecting sound waves that cause tympanic membrane **30** to vibrate. Cochlea **50** is spiral-shaped and filled with fluid. When sound vibrations

are transmitted from middle ear **40** to cochlea **50**, these sound waves cause the fluid inside cochlea **50** to move. This fluid movement can stimulate hair cells of cochlea **50** that convert mechanical sound waves into electrical signals sent to the brain of patient **10**.

[0043] Cochlea **50** forms round window niche **52**. Round window niche **52** generally includes a round window membrane. Round window niche **52** may represent an opening of cochlea **50** into middle ear **40**. Round window niche **52** and the round window membrane may act as a pressure relief valve for cochlea **50**, allowing for a displacement of fluid and the maintenance of proper pressure levels within cochlea **50**. When sound waves enter cochlea **50**, mechanical vibrations associated with the sound waves cause vibrations in the fluid, which stimulate the hair cells responsible for auditory perception. Round window niche **52** helps ensure that these mechanical vibrations occur without causing excessive pressure buildup within cochlea **50**, thus preserving a sensitivity of the auditory system.

[0044] Referring now to FIG. **2**, the patient **10** is depicted in an example suitable position and orientation to receive the procedure(s) to treat hearing loss and other ear disorders as described herein. In some cases, the procedure can be performed with the patient **10** fully supine as shown in FIG. **3** or reclined in a chair. The clinician **1** is configured to operate delivery tool **110** and treatment actuator device **130** to deliver the formulation to patient **10** while the patient **10** is in the example position illustrated in FIG. **2**.

[0045] For example, as shown in FIG. **2**, clinician **1** can grip delivery tool **110** with one hand and grip treatment actuator device **130** with another hand. Clinician **1** can proceed to advance delivery tool **110** into the ear of patient **10**. As described above, delivery tool **110** comprises side-by-side shafts **114**, **116** that advance simultaneously into the ear. The first shaft **114** can include a tip-mounted camera **112** on a distal tip that provides direct visualization of the second shaft **116** relative to anatomical features of the ear. This tip-mounted camera **112** can provide real-time image data via image data cable **142**. Display controller **140** receives the image data and outputs the image data for display by a screen of display device **150**. In some embodiments a latency between tip-mounted camera **112** capturing the image data and display device **150** displaying the image data is less than 10 milliseconds (ms). This means that by viewing the screen of display device **150** while operating delivery tool **110**, clinician **1** can see whether delivery tool **110** is placed to deliver the formulation to the targeted site in the ear of patient **10**.

[0046] With the direct visualization of second shaft **116** relative to anatomical features of the ear, clinician **1** can control treatment actuator device **130** to deliver the formulation to the targeted site when distal port **115** of second shaft **116** is located proximate to the targeted site such as round window niche **52**. The direct visualization provided by tip-mounted camera **112** can in some embodiments include a view of the formulation as it is being delivered and after it is delivered. This means that clinician **1** can view the real-time image data displayed by the screen of display device **150** to determine whether delivery tool **110** is placed to deliver the formulation and to determine whether the formulation is properly delivered at a location proximate the targeted site. When the formulation is delivered, clinician **1** can withdraw delivery tool **110** from the ear of patient **10**.

[0047] During a procedure to deliver the formulation, the head of the patient **10** can be rotated to between about 30 to 45 degrees away from the clinician **1** toward the opposite ear of the patient **10**. The jaw of the patient **10** can be slightly elevated, and/or the external portion of the ear of the patient **10** may be pulled superiorly and backward to adjust the canal aperture and angularity. As such, the round window niche **52** of the patient will be oriented generally upward (e.g., away from the ground) so that, upon dispensation of the formulation from the delivery tool, the formulation is able to pool at the round window niche **52** and not flow toward the eustachian tube or the ossicular chain.

[0048] In some implementations, the patient **10** remains awake during the procedure. That is, the procedure can be performed using a local anesthetic rather than a general anesthetic. For example,

in some cases agents such as phenol or lidocaine can be applied to the tympanic membrane **30** as a local anesthetic to facilitate the procedure. One reason that using local anesthetic instead of general anesthetic is possible is because in some embodiments delivery tool **110** is sized to traverse tympanic membrane **30** via a small needle puncture and does not require a larger incision to cross tympanic membrane **30**. Needle punctures can be minimally invasive and do not cause a great enough level of pain such that general anesthetic is necessary. In some cases, the patient **10** can be given general anesthesia for the procedure.

[0049] Referring to FIGS. **1-2**, a clinician **1** may grip delivery tool **110** with one hand and grip treatment actuator device **130** with another hand to deliver the formulation to the targeted site of patient **10**. Optionally, clinician **1** may grip delivery tool **110** with a dominant hand and grip treatment actuator device **130** with a non-dominant hand, but this is not required. As shown in FIG. **2**, the handle of delivery tool **110** has an axial length and gripping region that provides enhanced comfort when the clinician grasps it using a “pencil grip” (e.g., in a space between the thumb and the index finger) that achieves precise control over the movements. Delivery tool **110** is sized and shaped to be gripped equally effectively by the right hand or the left hand. Treatment actuator device **130** is also sized and shaped to be gripped equally effectively by the right hand or the left hand.

[0050] In this embodiment, the treatment actuator device **130** is a syringe device that includes a plunger actuator **131**, a fluid reservoir **132** (containing the therapeutic fluid), and a flexible tube **134** that extends distally toward a Luer connector at a proximal end of the delivery tool **110**. Thus, as depicted in FIGS. **1-3**, the clinician **1** can adjust the plunger actuator **131** to urge the therapeutic formulation along a fluid path extending between the fluid reservoir **132** and the distal port **115** at a distalmost end of the second shaft **116** of delivery tool **110**. Optionally, the fluid path is fully primed prior to advancement of the delivery tool **110** into the ear such that therapeutic fluid is already located in the second shaft **116** of the delivery tool **110** and ready for dispensation upon further actuation of the plunger actuator **131**.

[0051] In some embodiments, the plunger actuator **131** is configured to advance distally along a longitudinal axis of treatment actuator device **130**. As the plunger actuator **131** advances distally, a volume capacity of fluid reservoir **132** decreases. By decreasing the volume capacity of fluid reservoir **132**, plunger actuator **131** forces the formulation along the fluid path extending between the fluid reservoir **132** and the distal port **115**. As described above, the fluid path can be fully primed prior to advancement of the delivery tool **110** into the ear. This means that forcing formulation out of the fluid reservoir **132** displaces formulation already downstream in the fluid path to deliver the formulation to the targeted site via the distal port **115**.

[0052] As depicted in FIGS. **1-3**, the clinician **1** can monitor the direct visualization (via the tip-mounted camera in connection with the display device **150**) to identify when the distal port **115** of the delivery tool **110** is positioned to adjacent to or otherwise above (gravitationally) the targeted site of patient **10**, and then the clinician **1** can actuate the plunger actuator **131** of treatment actuator device **130** to urge delivery of the therapeutic formation from the distal port **115** for deposition at the targeted site. **134**

[0053] In some cases, treatment actuator device **130** is not rigidly attached to the delivery tool **110**, thereby permitting the clinician **1** to accurately maintain a stationary position of the delivery tool **110** even when applying a force or other manual manipulation to the treatment actuator device **130**. For example, delivery tool **110** and treatment actuator device **130** are attached to either end of flexible tube **134**. Delivery tool **110** and treatment actuator device **130** can be positioned so that there is slack in flexible tube **134**. When there is slack in flexible tube **134**, treatment actuator device **130** can move without displacing delivery tool **110** from a stationary position. This improves an ability of clinician **1** to operate delivery tool **110** using one hand and treatment actuator device **130** using another as compared with medical device systems that include a treatment actuator device rigidly attached to a delivery tool.

[0054] As previously described, in this embodiment, the flexible tube **134** of the treatment actuator device **130** includes a Luer lock for removably mating with a Luer connector mounted to the proximal end of delivery tool **110**. For example, the distal end of the flexible tube **134** can include a first (female) fitting with internal threads, and the Luer connector at the proximal end of the delivery tool **110** can include a second (male) fitting with external threads to match the internal threads of the first fitting. The first fitting receives the second fitting such that the external threads engage with the internal threads to form a sealed connection. This tight seal prevents leakage of fluid from the Luer lock connection. The connection between flexible tube **134** is not limited to examples where flexible tube **134** includes a male and delivery tool **110** includes a female fitting. In some embodiments, flexible tube **134** includes a female fitting and delivery tool **110** includes a male fitting.

[0055] It should be understood from the description herein that, in other embodiments, the delivery tool **110** and the flexible tube **134** are not limited to a Luer lock connection. For example, the delivery tool **110** and the flexible tube **134** can be connected to one another using other threaded connections, snap-fit connections, barbed connections, compression fittings, adhesive bonding, welding or fusion bonding, and bayonet connections. In any case, delivery tool **110** and flexible tube **134** are connected so that a lumen of flexible tube **134** and a lumen of delivery tool **110** form a single fluid path for delivering a formulation to a targeted site in the ear of patient **10**.

[0056] Referring now to FIG. **3**, the delivery tool **110** can be equipped with a tip-mounted camera **112** configured to provide real-time imaging of a second shaft **116** and surrounding anatomy within the ear. Distal port **115** located at distalmost end of second shaft **116** may, in some examples, deliver a formulation **160** to the targeted site of patient **10**. This means that tip-mounted camera **112** can provide direct visualization by capturing the location of the distalmost end of second shaft **116** relative to one or more anatomical regions of the ear of patient **10**. This allows clinician **1** to control the treatment actuator device **130** to deliver the formulation **160** at a select time when the distal port **115** of delivery tool **110** is accurately positioned proximate the targeted site within the ear of patient **10**. As described in more detail below, the tip-mounted camera **112** can include an image capture device such as a CMOS image sensor that provides a camera-on-a-chip construction having a width of less than 1.5 mm, and preferably 1 mm or less) mounted at a distal end of second shaft **116** of the delivery tool **110**.

[0057] Such a CMOS camera mounted at the tip of the delivery tool **110** can advantageously reduce the overall combined size of the shafts **114**, **116** that pass through the tympanic membrane **30** as compared with tools that use an endoscope to provide direct visualization (FIG. **3**). This is because an endoscope lens mounted at the end of an endoscope cannula can have a diameter that is greater than a width of the image sensor of the CMOS camera mounted on the distal tip of the first shaft **114**. This means that including a CMOS camera at the distal tip of the first shaft **114** instead of an endoscope lens can result in first shaft **114** having a diameter that is smaller than a width of a comparable endoscope cannula. Tip-mounted camera **112** is not limited to including a CMOS camera. In some embodiments, tip-mounted camera **112** can include an image capture device other than a CMOS camera. For example, in some alternative options, tip-mounted camera **112** can include an optical coherence tomography (OCT) image capture device.

[0058] Optionally, the delivery tool **110** also includes a light-emitting diode (LED) mounted on a distal tip of first shaft **114** adjacent to the CMOS camera to provide illumination within the middle ear toward the distal port of the delivery tool **110**. In such embodiments, the CMOS camera may include an image sensor comprising an array of light-sensitive pixels arranged on a semiconductor chip. When light reaches the image sensor, each pixel converts the light into an electrical signal proportional to an intensity of the light at the pixel. The sensor captures an image indicating the intensity at each pixel. Optionally, tip-mounted camera **112** includes a lens to focus light onto the image sensor of the CMOS camera.

[0059] Tip-mounted camera **112** can include circuitry for generating image data based on optical

signals received by the image sensor. The circuitry configured to generate the image data may, in some cases, be part of the image sensor of the CMOS camera mounted at the distal tip of the delivery tool **110**. In other cases, the circuitry configured to generate the image data is separate from the image sensor. In some examples, to generate the image data, the circuitry generates, for each image frame of a sequence of image frames, an intensity value for each pixel of the array of light-sensitive pixels.

[0060] Tip-mounted camera **112** can, in some implementations, output image data to display controller **140** via image data cable **142**. Display controller **140** can be configured to connect with the image data cable **142** and convert the image data to another output, such as a USB, HDMI or display port connection that is then output the display device **150**. In some examples, display controller **140** causes display device **150** to display the image data in real time so that clinician **1** has real time visualization of the location of delivery tool **110** relative to anatomical landmarks within the ear of patient **10**. This allows clinician **1** to determine whether delivery tool **110** is properly positioned prior to dispensing the therapeutic formulation **160** to the targeted site of patient **10**. In some examples, display controller **140** can include multiple output ports for connection to a variety of different types of display devices **150**, such as connections to television, a computer monitor, a smart phone, a laptop, a or a tablet computer. In some examples, display controller **140** includes an input port that can connect to and disconnect from image data cable **142**. Display controller **140** can connect to and disconnect from more than one delivery tool **110**.

[0061] Still referring to FIG. **3**, distal portion **111** of delivery tool **110** can be inserted through tympanic membrane **30** into middle ear **40**. Distal portion **111** includes a portion of first shaft **114** that includes a tip-mounted camera **112** and a portion second shaft **116** that includes distal port **115** configured to deliver the formulation **160** to the targeted site. This means that distal port **115** can be positioned to deliver the formulation **160** to a targeted site at round window niche **52** of cochlea **50** as tip-mounted camera **112** provides direct visualization. Tip-mounted camera **112** can output image data indicating the location of the delivery tool **110** relative to anatomical features of the ear of patient **10** such as the round window niche **52**. Based on this relative location of the distal portion of delivery tool **110**, clinician **1** can operate plunger actuator **131** of treatment actuator device **130** to deliver the formulation **160**.

[0062] The therapeutic formulation **160** that is deposited into the round window niche **52** of the cochlea **50** can include at least one active agent configured to transfer passively by diffusion across the round window membrane of the round window niche **52**, according to a concentration gradient, and into the perilymph within the cochlea **50**. Therapeutic formulation **160** can be a self-gelling material. As such, the formulation **160** that is delivered adjacent to the round window membrane can thereafter reside adjacent to or within the round window niche **52** as a semi-solid gel substance. As a gel substance, the delivery of the formulation **160** will remain in the targeted site at the cochlea **50** so that the formulation **160** can gradually release its active ingredient for an extended period of time such as days, weeks, or even months. Formulation **160** may ensure stability, bioavailability, and compatibility with the delicate structures of the inner ear.

[0063] After the delivery of the therapeutic formulation **160**, the delivery tool **110** can be removed from the patient **10**. The therapeutic formulation **160** (e.g., in gel form) will remain at the targeted site in the cochlea **50** to provide extended therapeutic effects by a controlled, sustained release of the active ingredient into the body of the patient **10**. Sustained release can encompass the release of effective amounts of an active ingredient of the formulation **160** for an extended period of time. The sustained release may encompass first order release of the active ingredient, zero order release of the active ingredient, or other kinetics of release such as intermediate to zero order and first order, or combinations thereof. The sustained release may also encompass controlled release of the active ingredient of the formulation via passive molecular diffusion driven by a concentration gradient across a membrane or porous structure.

[0064] When delivery tool **110** delivers formulation **160**, formulation **160** can be absorbed through

the round window membrane and distributed within cochlea **50**, where the formulation exerts therapeutic effects. The formulation may target specific structures within the cochlea, such as hair cells or sensory neurons, depending on the nature of the treatment. Once inside the cochlea **50**, formulation **160** can treat a range of conditions or symptoms, including sensorineural hearing loss, tinnitus, or inner ear disorders. By delivering formulation **160** directly to the round window membrane, delivery tool **110** can provide trans-tympanic membrane therapy that achieves higher local concentrations and enhances therapeutic outcomes as compared with systems that do not deliver therapy directly to the round window membrane.

[0065] The procedure for delivering the formulation into the cochlea **50** of the patient **10** can be repeated periodically as needed for a particular patient's treatment. For example, in some cases deliveries of the formulation can be administered about every three to 24 months, each time using new delivery tools as described herein. In particular cases, an assessment of the patient **10** can be performed to determine whether or when to administer more formulation. In some cases, a procedure such as magnetic resonance imaging (MRI) (or other type of procedure) can be performed to help make such an assessment.

[0066] Referring now to FIGS. **4-6**, an example delivery tool **110** can be used to perform the procedure to treat hearing loss and other ear disorders as described herein. As illustrated in FIGS. **4-6**, delivery tool **110** includes tip-mounted camera **112**, first shaft **114**, second shaft **116**, and handle **118**. Handle **118** receives first shaft **114** and second shaft **116** through a distal opening **120** of handle **118**.

[0067] As described above, delivery tool **110** is configured to deliver a formulation to a targeted site in an ear of patient **10**. A portion of delivery tool **110** is inserted into the ear of patient **10** to deliver the formulation. Since the human ear forms several small caverns, passageways, and openings, delivery tool **110** can be sized to access these areas and successfully deliver the formulation. Delivery tool **110** is also sized to be gripped and maneuvered by clinician **1** during a procedure to deliver the formulation. Physical dimensions and aspects of delivery tool **110** allow delivery tool **110** to advance into the ear of patient **10** to deliver the formulation to the targeted site through a minimally invasive approach.

[0068] Delivery tool **110** can provide trans-tympanic membrane access to middle ear **40** and round window niche **52**. Trans-tympanic membrane therapy delivery to middle ear **40**, specifically targeting the round window membrane, can involve administering medication or therapeutic agents directly into middle ear **40** through round window niche **52** of cochlea **50**. Delivery tool **110** can access middle ear **40** via a small puncture in tympanic membrane **30** while patient **10** is under local anesthesia. In some implementations, a size of this small puncture is more similar to a size of a needle puncture than to a size of a surgical incision made with a scalpel. This means that when delivery tool **110** accesses middle ear **40** through the small puncture in tympanic membrane **30**, delivery tool **110** is sized to cross the tympanic membrane **30** through the small puncture without significantly increasing the size of the small puncture and without further damaging tympanic membrane **30**.

[0069] For example, distal portion **111** of delivery tool **110** can extend through the small puncture in tympanic membrane **30** and into the middle ear **40**. In some embodiments, a greatest width of distal portion **111** that extends into the middle ear **40** is less than 3 mm, preferably less than 2 mm, and within a range from 1.0 mm to 1.8 mm. This means that distal portion **111** of delivery tool **110** can enter the small puncture in tympanic membrane **30** without significantly increasing a size of the small puncture, ensuring that the procedure is minimally invasive. In some embodiments, the greatest width of distal portion **111** is smaller than a greatest width of a puncture tool that creates the small puncture in tympanic membrane **30**. In some embodiments, the greatest width of distal portion **111** is less than 150% of a greatest width of a puncture tool that creates the small puncture in tympanic membrane **30**.

[0070] Optionally, the distal portion **111** of the delivery tool **110** can be used as the puncture tool to

achieve the trans-tympanic access (e.g., without the need for a separate puncture tool). In some embodiments, delivery tool **110** can create the small puncture in tympanic membrane **30** itself by using blunt dissection to pierce tympanic membrane **30**. For example, the user can readily grasp the handle **118** (e.g., using a pencil grip or other grip that enhances control and accuracy) and advance the side-by-side shafts **114** and **116** into the outer ear toward the tympanic membrane **30**. Because the distal end of the second shaft **116** extends distally of the first shaft **114** and has a small diameter (for example, less than 1.5 mm and preferably about 1.3 mm in this embodiment), the distal end of the second shaft **116** can engage the tympanic membrane to create a small pilot opening through the tympanic membrane **30**. The tip-mounted camera **112** on the first shaft **114** can provide direct visualization of the distal end of the second shaft **116** during formation of the pilot opening through the tympanic membrane **30**. As the handle **118** is manipulated to further advance both shafts **114** and **116**, the first shaft **114** then reaches the tympanic membrane **30** and passes through the pilot opening in the tympanic membrane previously formed by the second shaft **116** (which can cause the pilot opening to dilate slightly). As such, the distal portion **11** of the delivery tool **110** can be used to safely and rapidly achieve a micro-puncture in the tympanic membrane **30** that is capable of self-healing, with the overall size of the opening in the tympanic membrane **30** being substantially the same size as the combined lateral width of the side-by-side shafts. In such examples, the opening formed in the tympanic membrane can be no greater than 2 mm, and preferably about 1 mm to about 1.8 mm.

[0071] Tip-mounted camera **112** provides direct visualization of a location of the distal end of the delivery tool **110** relative anatomical features of the ear of patient **10** during the procedure to treat hearing loss and other ear disorders as described herein. To provide direct visualization, an image sensor of tip-mounted camera **112** is located on the distal portion **111** of delivery tool **110** that extends through the small puncture in tympanic membrane **30** and into middle ear **40**. This direct visualization provided by tip-mounted camera **112** allows clinician **1** to determine whether delivery tool **110** is properly positioned to deliver a formulation to the targeted site.

[0072] First shaft **114** comprises an elongated member that extends from a proximal end within handle **118** to a distal end **122**. Preferably, first shaft **114** is cylindrical and has a circular cross section along an entire length of first shaft **114**, but this is not required. Optionally, first shaft **114** can form a shape other than a cylinder having a circular cross section. In some embodiments, an image sensor of tip-mounted camera **112** can be mounted on the distal end **122** of first shaft **114**. In some embodiments, a light source of tip-mounted camera **112** such as an LED is mounted on the distal end **122** of first shaft **114**.

[0073] Second shaft **116** comprises an elongated member that extends from a proximal end within handle **118** to a distal end **124**. Preferably, second shaft **116** is cylindrical and has a circular cross section along an entire length of second shaft **116**, but this is not required. Second shaft **116** can form a shape other than a cylinder. In some embodiments, second shaft **116** forms a lumen that extends an entire length of second shaft **116** from a proximal end of second shaft **116** to distal end **124** of second shaft **116**. Second shaft **116** defines a distal port **115** for delivering the formulation to the targeted site via the lumen of second shaft **116**.

[0074] Shafts **114**, **116** are sized and positioned so that delivery tool **110** can effectively deliver the formulation to the targeted site as tip-mounted camera **112** provides direct visualization. For example, first shaft **114** and second shaft **116** can be fixedly attached to each other so that first shaft **114** and second shaft **116** cannot move relative to each other. Tip-mounted camera **112** can be fixedly attached to a distal portion of first shaft **114**. Since first shaft **114** and second shaft **116** can be fixedly attached, this means that the image sensor of tip-mounted camera **112** does not move relative to the location of first shaft **114** and the location of tip-mounted camera **112** does not move relative to second shaft **116** in some embodiments.

[0075] Fixedly attaching first shaft **114** and second shaft **116** provides several advantages over tools that include separate delivery and imaging shafts that move relative to each other. One

advantage is that a clinician can move first shaft **114** and second shaft **116** simultaneously without controlling the position of second shaft **116** relative to the position of first shaft **114**. Another advantage is that because the location of tip-mounted camera **112** is fixed relative to second shaft **116**, tip-mounted camera **112** can provide direct visualization of the location of second shaft **116** relative to anatomy of the ear. When first shaft **114** and second shaft **116** are fixed, direct visualization is not interrupted by tip-mounted camera **112** moving distally away from the distal end of second shaft **116** so that the distal end of second shaft **116** is not in focus or moving proximally past the end of second shaft **116** so that the distal end of second shaft **116** is behind the tip-mounted camera **112**.

[0076] In some embodiments, first shaft **114** and second shaft **116** are rigid and not bendable. A passageway through external ear canal **38** and across tympanic membrane **30** is generally straight in many patients so that delivery tool **110** can advance to the targeted site without bending. This means that first shaft **114** and second shaft **116** can be generally straight members that extend along a longitudinal axis without bending relative to the longitudinal axis of delivery tool **110** during use. It can be beneficial for first shaft **114** and second shaft **116** to be rigid so that clinician **1** can operate delivery tool **110** without independently controlling first shaft **114** and second shaft **116** to bend.

[0077] Distal end **124** of second shaft **116** extends distally beyond distal end **122** of first shaft **114**. This configuration helps delivery tool **110** to provide direct visualization of the location of second shaft **116** relative to anatomy of the ear. For example, tip-mounted camera **112** is placed on the distal end **122** of first shaft **114**. This means that tip-mounted camera **112** can capture a portion of second shaft **116** that extends distally beyond distal end **122** of shaft **114**. Delivery tool **110** can deliver a formulation to a targeted site via distal port **115** at the distal end **124** of second shaft **116**. The portion of second shaft **116** that extends distally beyond distal end **122** of shaft **114** is within a field of view of tip-mounted camera **112**. When distal port **115** is positioned adjacent to the targeted site, delivery tool **110** delivers the formulation to the targeted site via distal port **115**.

[0078] First shaft **114** extends from distal opening **120** of handle **118** for a distance equal to length **L1**. Second shaft **116** extends from distal opening **120** of handle **118** for a distance equal to length **L2**. As shown in FIG. 4, length **L2** is greater than length **L1**, meaning that the distal end **124** of second shaft **116** extends distally beyond distal end **122** of first shaft **114**. The distance between the distal end **124** of second shaft **116** and the distal end **122** of first shaft **114** is equal to length **L3**.

[0079] Delivery tool **110** can be sized so that lengths **L1**, **L2**, and **L3** help delivery tool **110** to perform a procedure to deliver the formulation to a targeted site within the middle ear **40** of patient **10**. For example, the tympanic membrane **30** is located at an end of external ear canal **38**. This means that delivery tool **110** can be sized to navigate through ear canal **38** to reach middle ear **40**, located across tympanic membrane **30** at the end of external ear canal **38**. In some embodiments, delivery tool **110** can be sized so that tip-mounted camera **112** and distal port **115** both reach middle ear **40** when first shaft **114** and second shaft **116** simultaneously advance through tympanic membrane **30** via a needle puncture.

[0080] A length of the external ear canal for many adult patients is within a range from 20 mm to 30 mm. In some embodiments, delivery tool **110** can be sized so that exposed portions of shafts **114**, **116** extend through most of the external ear canal of many patients. This is because the external ear canal is a narrow passageway. As shown in FIGS. 4-6, a combined width of shafts **114**, **116** is significantly narrower than a width of handle **118**. This means that when delivery tool **110** is sized so that the exposed portions of shafts **114**, **116** extend through most of the external ear canal. This can ensure that delivery tool **110** is able to navigate through the external ear canal to the targeted site without wider portions of delivery tool **110** being physically obstructed by walls of the external ear canal or other anatomy of the ear.

[0081] In some embodiments, the length **L1** of the portion of first shaft **114** extending distally from distal opening **120** of handle **118** is within a range from 20 mm to 30 mm. Preferably, length **L1** is equal to approximately 25 mm. The length **L2** of the portion of second shaft **116** extending distally

from distal opening **120** of handle **118** may be within a range from 22.5 mm to 33.5 mm. Preferably, length L2 is equal to approximately 27.5 mm. Length L2 is preferably greater than length L1. This means that delivery tool **110** can be sized so that the portions of shafts **114**, **116** that extend from distal opening **120** of handle **118** are similar to a length of the external ear canal of many patients.

[0082] Length L3 represents a distance between distal end **122** of first shaft **114** and distal end **124** of second shaft **116**. As described above, tip-mounted camera **112** can be located on the distal end **122** of first shaft **114** so that tip-mounted camera **112** captures a portion of second shaft **116** that extends distally beyond the distal end **122** of first shaft **114**. In some embodiments, delivery tool **110** is sized to perform one or more tasks within middle ear **40**. For example, length L3 can be sized so that tip-mounted camera **112** and distal port **115** both fit within middle ear **40** at the same time. This allows tip-mounted camera **112** to capture the location of second shaft **116** relative to a targeted site such as round window niche **52**. The length L3 can also be sized so a distalmost portion of second shaft **116** is within a field of view of tip-mounted camera **112**.

[0083] In some embodiments, length L3 is within a range from 1 mm to 4 mm and preferably about 2.5 mm. When length L3 is about 2.5 mm, for example, first shaft **114** and second shaft **116** can advance simultaneously across tympanic membrane **30** so that tip-mounted camera **112** and distal port **115** are both within middle ear **40** at the same time. Since tip-mounted camera **112** is located on a distal end **122** of first shaft **114**, tip-mounted camera **112** is configured to capture the distalmost portion of second shaft **116** to provide direct visualization of second shaft **116** relative to the targeted site beyond tympanic membrane **30**. When the targeted site is the round window membrane, for example, delivery tool **110** having a length L3 between distal end **122** of first shaft **114** and distal end **124** of second shaft **116** allows shafts **114**, **116** to move simultaneously so that distal port **115** is located adjacent the round window membrane while tip-mounted camera **112** is located within middle ear **40** to capture both second shaft **116** and the round window membrane.

[0084] Delivery tool **110** can be sized so that length L3 is compatible with a size of the middle ear **40** for many patients. For example, delivery tool **110** can be sized so that length L3 is less than a distance between the tympanic membrane **30** and round window niche **52**. When length L3 is less than a distance between the tympanic membrane **30** and round window niche **52**, this allows delivery tool **110** to advance so that tip-mounted camera **112** advances beyond tympanic membrane **30** and so that distal port **115** advances to a location adjacent round window niche **52**.

[0085] Delivery tool **110** having shafts **114**, **116** that are sized to length L1 and the length L2 allows delivery tool **110** to navigate through external ear canal **38** and traverse tympanic membrane **30** without wider portions of delivery tool **110** being obstructed by walls of the external ear canal **38** or other anatomy of the ear. For example, when the portion of first shaft **114** extending distally from distal opening **120** of handle **118** has length L1 and when the portion of second shaft **116** extending distally from distal opening **120** of handle **118** has length L2, delivery tool **110** can advance to the targeted site in the ear of patient **10** such that first shaft **114**, second shaft **116**, and a distal section **136** of handle **118** are within the ear of patient **10**. That is, when length L1 is within a range from 20 mm to 30 mm and length L2 is within a range from 22.5 mm to 33.5 mm, the distal section **136** of delivery tool **110** can be inserted into the ear and the proximal section **138** of delivery tool **110** can remain outside of the ear.

[0086] First shaft **114** and second shaft **116** are positioned to allow delivery tool **110** easy access to and maneuverability within small locations of the ear of patient **10** such as external ear canal **38** and middle ear **40**. In some embodiments, first shaft **114** and second shaft **116** are in contact along an entire length of first shaft **114** extending distally from handle **118**. This means that along an entire length of first shaft **114** that extends distally from handle **118**, a width of delivery tool **110** is equal to a sum of a width of first shaft **114** and a width of second shaft **116**. In other words, there is no point along first shaft **114** extending distally from handle **118** where first shaft **114** and second shaft **116** are separated by a gap.

[0087] When there is no gap between first shaft **114** and a width of second shaft **116**, this minimizes cross-sectional footprint of first shaft **114** and a width of second shaft **116** and makes first shaft **114** and second shaft **116** suited for a minimally invasive procedure to deliver a formulation. This is because first shaft **114** and second shaft **116** can simultaneously advance through a small needle puncture in tympanic membrane **30** when first shaft **114** and second shaft **116** are grouped closely together so that there is no space between first shaft **114** and second shaft **116**. Any space between first shaft **114** and second shaft **116** could make it harder for first shaft **114** and second shaft **116** to simultaneously advance through the same needle puncture.

[0088] Optionally, first shaft **114** and second shaft **116** can be separated by a small gap along an entire length of first shaft **114** extending distally from handle **118**. In some examples, a width of this gap may be within a range from 0.1% to 2% of an outer diameter of first shaft **114**. When a gap between first shaft **114** and second shaft **116** is small relative to a diameter of first shaft **114**, this gap does not significantly add to a total cross-sectional footprint of first shaft **114** and second shaft **116**. This means that first shaft **114** and second shaft **116** can advance through a needle puncture in tympanic membrane **30** in some implementations where there is a gap between first shaft **114** and second shaft **116**.

[0089] As shown in FIGS. 4-6, first shaft **114** and second shaft **116** are parallel along an entire length of first shaft **114** and second shaft **116** extending distally from handle **118**. Since first shaft **114** and second shaft **116** are parallel, a total distance across first shaft **114** and second shaft **116** does not change along the length of first shaft **114** extending distally from handle **118**. This means that delivery tool **110** can advance through the external ear canal **38** and middle ear **40** ear of patient **10** without being obstructed by the walls of external ear canal **38** or other ear anatomy. A tool having non-parallel shafts includes at least some portions where there is a gap between the non-parallel shafts. These gaps increase the total width of the tool, making it more difficult to navigate the tool in small spaces.

[0090] Since first shaft **114** and second shaft **116** are parallel, this may allow delivery tool **110** to fit within the ear more easily as compared with tools that use two shafts that are not parallel. When two shafts are not parallel, these shafts may intersect at an intersection point and grow farther apart from each other along an axis. When shafts are farther apart, this may increase a diameter of a passageway needed to accommodate the shafts. But when the shafts are parallel, this may improve an ability of the shafts to within a narrow passageway such as external ear canal **38**.

[0091] In some embodiments, first shaft **114** and second shaft **116** are not exactly parallel, and there is a small angle between a first shaft **114** and second shaft **116**. In some embodiments, this angle is less than 1 degree. When the angle between first shaft **114** and second shaft **116** is small and first shaft **114** and second shaft **116** are close together, a maximum width of a gap between first shaft **114** and second shaft **116** does not significantly add to a total width of the first shaft **114** and second shaft **116** extending distally from handle **118**. This allows delivery tool **110** to navigate to the targeted site.

[0092] In some embodiments, one or both of first shaft **114** and second shaft **116** comprise a polyimide material. Polyimide is a versatile polymer that exhibits high temperature resistance, mechanical strength, electrical insulation, and biocompatibility. This makes polyimide suitable for medical applications where flexibility, durability, and compatibility with the human body are beneficial. Polyimide can withstand high temperatures without significant degradation. Polyimide also exhibits mechanical properties such as high tensile strength, stiffness, and dimensional stability. Since polyimide provides electrical insulation properties, it may be beneficial for first shaft **114** to include polyimide for insulating an electrical conductor connected to tip-mounted camera **112**. Since polyimide has a low coefficient of thermal expansion, it may be beneficial to use polyimide for first shaft **114** and second shaft **116** so that body heat does not change a shape and size of first shaft **114** and second shaft **116**. Furthermore, many formulations of polyimide are biocompatible, meaning that polyimide is safe for use within the human body, such as within the

ear of patient **10**. This property makes polyimide suitable to use as part of first shaft **114** and second shaft **116**, since first shaft **114** and second shaft **116** are inserted into the ear of patient **10**. [0093] In some embodiments, one or both of first shaft **114** and second shaft **116** include a stainless steel material such as stainless steel hypodermic tubing. Stainless steel hypodermic tubing is a kind of hollow tube made from stainless steel. In some cases, hypodermic tubing can be manufactured to precise dimensions (e.g., inner diameter, outer diameter, length) to ensure consistency in diameter, wall thickness, and surface finish. This precision is crucial for applications such as medical devices and instruments. Stainless steel resists corrosion, making this material useful in environments such as the human ear where moisture, chemicals, and bodily fluids are located. Stainless steel can be strong and durable, which allows stainless steel hypodermic tubing to withstand high pressures and mechanical stresses. Stainless steel can also be biocompatible, meaning that stainless steel is well-tolerated by the human body without causing adverse reactions.

[0094] First shaft **114** and second shaft **116** are not limited to including polyimide and/or stainless steel. In some embodiments, one or both of first shaft **114** and second shaft **116** include materials such as titanium, platinum, Nitinol, Cobalt-Chromium alloys, polyethylene, polyethylene terephthalate (PET), polyetheretherketone (PEEK), ceramics, fluoropolymers, silicone, or any combination thereof. In any case, first shaft **114** and second shaft **116** may comprise materials that are biocompatible with the human ear and possess mechanical and thermal properties suitable for delivering the formulation to the targeted site within the ear of patient **10**.

[0095] In some embodiments, first shaft **114** has an outer diameter (refer to D1 in FIG. **10** below) within a range from 0.5 mm to 1.5 mm, and preferably from 1.0 mm to 1.4 mm. In one embodiment, first shaft **114** has an outer diameter of 1.3 mm. It may be beneficial for first shaft **114** to have a small outer diameter that accommodates tip-mounted camera **112** so that the profile of first shaft **114** through tympanic membrane **30** is limited. For example, when tip-mounted camera **112** includes a CMOS camera and an LED, this may allow first shaft **114** to have a small outer diameter within the range from 0.5 mm to 1.5 mm. When tip-mounted camera **112** includes a light source removed from the tip of first shaft **114** and an optical waveguide to bring light from the light source to the tip of first shaft **114**, this may decrease a diameter of first shaft **114** as compared with embodiments where a light source (e.g., an LED) is located at the tip of first shaft **114**. In any case, an outer diameter of first shaft **114** may allow first shaft **114** to traverse tympanic membrane **30** in a minimally invasive procedure to deliver a formulation to a targeted site in an ear of patient **10**.

[0096] Second shaft **116**, in some embodiments, has an outer diameter (refer to D2 in FIG. **10** below) within a range from 0.20 mm to 0.50 mm. Preferably, the outer diameter of second shaft **116** is 0.41 mm. When the outer diameter of second shaft **116** is within a range from 0.20 mm to 0.50 mm, this may limit a profile of second shaft **116** through tympanic membrane **30** so that delivering the formulation to the targeted site in the ear of patient **10** is minimally invasive. That is, when the outer diameter of second shaft **116** is within a range from 0.20 mm to 0.50 mm, the profile of second shaft **116** is more like a needle than a catheter, thus limiting traumatic impact to the tympanic membrane **30**. In an embodiment, where an outer diameter of first shaft **114** is 1.3 mm and an outer diameter of second shaft **116** is 0.41 mm, a total distance across first shaft **114** and second shaft **116** is 1.66 mm. This width allows first shaft **114** and second shaft **116** to simultaneously advance through a needle puncture in tympanic membrane **30** while minimizing trauma to the tympanic membrane **30**.

[0097] In some embodiments, one or both of a tip region of first shaft **114** and a tip region of second shaft **116** comprises a silicone material. Silicone is generally less rigid than polyimide. Polyimide is a high-performance polymer known for its excellent mechanical strength, dimensional stability, and high temperature resistance and is often used in applications where rigidity and stability are critical. Silicone is a flexible and elastomeric material. While silicone can vary in hardness depending on formulation and processing, silicone is typically softer and more flexible compared to polyimide. This means that it may be beneficial to use silicone at one or both of a tip

region of first shaft **114** and a tip region of second shaft **116**, because the tip regions of first shaft **114** and second shaft **116** cross the tympanic membrane **30** of patient **10**. In other words, first shaft **114** and second shaft **116** can be less traumatic to the ear of patient **10** when first shaft **114** and second shaft **116** comprise silicone tips as compared with examples where shafts do not include silicone tips.

[0098] Delivery tool **110** includes handle **118** configured to receive a proximal portion of first shaft **114** and a proximal portion of second shaft **116**. In some examples, handle **118** receives the first shaft **114** and the second shaft **116** through distal opening **120** of handle **118**. Handle **118** can be sized so that a clinician is able to firmly grasp handle **118** with one hand while simultaneously advancing first shaft **114** and second shaft **116** of delivery tool **110** toward a targeted site. First shaft **114** and second shaft **116** can be fixedly attached to handle **118** so that first shaft **114** does not move relative to handle **118** and second shaft **116** does not move relative to handle **118**. First shaft **114** and second shaft **116** can extend distally from handle **118** via distal opening **120**.

[0099] In some embodiments, a distal section **136** of delivery tool **110** can be inserted into the ear of patient **10** and a proximal section **138** of delivery tool **110** can remain outside of the ear of patient **10** during a procedure to deliver a formulation. In other words, the clinician can insert delivery tool **110** up to a point on handle **118** that is midway between distal opening **120** of handle **118** and grip edges **135** of handle **118**. Distal section **136** of delivery tool **110** is generally narrower than proximal section **138** of delivery tool **110**. Distal section **136** can be sized to fit within the ear of patient **10** so that delivery tool **110** can advance to the targeted site within the ear of patient **10**. Proximal section **138** can be sized for gripping by a user such as clinician **1**. This means that a width of proximal section **138** can be sized to rest firmly and comfortably within features and contours of the human hand.

[0100] As shown in FIGS. **4-6**, a surface of handle **118** is curvilinear in shape. For example, the surface of handle **118** is curved such that some portions of handle **118** have a diameter that is greater than other portions of handle **118**. In some embodiments, these curves can improve an ability of a clinician to grip handle **118** as compared with tools that do not have curvilinear surfaces in a grip area. This is because peaks and valleys can be easier to hold than flat surfaces. The peaks and valleys of handle **118** can engage with features of the clinician's hand, thus preventing handle **118** from slipping.

[0101] In some embodiments, handle **118** includes grip edges **135** on a surface of handle **118**. Grip edges **135** can improve an ability of clinician **1** to grip handle **118** as compared with an ability of clinician **1** to grip a handle that does not include grip edges. This is because grip edges **135** can increase a coefficient of friction between fingers of clinician **1** and the surface of handle **118** as compared with a coefficient of friction between fingers and a smooth surface. The increased coefficient of friction leads to a decreased likelihood that handle **118** will slip from the grip of clinician **1**. As shown in the illustrated embodiment, grip edges **135** are located on a part of handle **118** that is between a midpoint of handle **118** and a distal end of handle **118**.

[0102] As shown in FIGS. **4-6**, the surface of handle **118** forms an undulating pattern. This undulating pattern gradually increases from a proximal end **144** of handle **118** to a first peak **145** of the surface of handle **118**. The undulating pattern gradually decreases from a first peak **145** to a valley **146**. The undulating pattern gradually increases again from a valley **146** to a second peak **147**. From the second peak **147** to the distal opening **120** of handle **118**, the undulating pattern of the surface gradually decreases.

[0103] In some examples, a cross-section of handle **118** at each point along a length of handle **118** is round (e.g., a circle, an oval). This means that a diameter of a cross-section of handle **118** at first peak **145** can be greater than a diameter of the cross-section of handle **118** at valley **146** and a diameter of a cross-section of handle **118** at second peak **147** is greater than a diameter of the cross-section of handle **118** at valley **146**. A clinician can grip delivery tool **110** using a pencil grip. When delivery tool **110** is gripped using a pencil grip, a portion of handle **118** at valley **146** can rest in a

space between the thumb and the index finger while the thumb, the index finger, and the middle finger engage hold grip edges **135**.

[0104] Delivery tool **110** includes a connector device **151** attached to a proximal end **144** of handle **118**. As described above, connector device **151** may represent a Luer connector for mating with a Luer lock of flexible tube **134** of FIGS. **1-2**. This means that delivery tool **110** is configured to receive the formulation through connector device **151** located at the proximal end **144** of handle **118**.

[0105] FIG. **4** illustrates a first view of delivery tool **110** from a first perspective looking at a side of delivery tool **110**, where first shaft **114** is arranged above second shaft **116** in the example of FIG. **5**. FIG. **5** illustrates a second view of delivery tool **110** from a second perspective looking at a side of delivery tool **110**, where first shaft **114** and second shaft **116** are aligned in the same plane extending into the page. This means that the delivery tool **110** in the second view of FIG. **5** is rotated 90 degrees relative to the delivery tool **110** in the first view of FIG. **4**. FIG. **6** illustrates a cutaway view of delivery tool **110** from the first perspective looking at the side of delivery tool **110**.

[0106] Referring now to FIG. **6**, tip-mounted camera **112** can, in some embodiments, be connected to an electrical conductor **113**. As shown in FIG. **6**, electrical conductor **113** can extend through a lumen **148** formed by first shaft **114**. In some embodiments, electrical conductor **113** may extend from tip-mounted camera **112** out of delivery tool **110** through a gap between a proximal end of handle **118** and connector device **151**. Electrical conductor **113** may, in some examples, connect to display controller **140** of FIG. **1**. For example, electrical conductor **113** can be disposed within image data cable **142** in some embodiments. Delivery tool **110** is not limited to the embodiment illustrated in FIG. **6**. In some examples, electrical conductor **113** may exit delivery tool **110** at another location such as the proximal end of connector device **250**, a location between a proximal end of handle **118** and a distal end of handle **118**, or another location.

[0107] Electrical conductor **113** can provide power to one or more components of tip-mounted camera **112** and/or facilitate communication with one or more components of tip-mounted camera **112** and other devices. For example, electrical conductor **113** may provide power to a camera sensor and/or a light source of tip-mounted camera **112**. Additionally, or alternatively, a camera source of tip-mounted camera **112** can transmit image data via electrical conductor **113**. Electrical conductor **113** is not limited to including a single conductor. In some embodiments, electrical conductor **113** includes more than one conductor, such as one conductor connected to an LED and another conductor connected to a camera sensor.

[0108] Delivery tool **110** is configured to deliver the formulation via the lumen **149** formed by second shaft **116** so that the formulation exits the distal opening at distal end **124** of second shaft **116**. As shown in FIG. **6**, distal port **115** is located at a distal end of lumen **149** that is defined by second shaft **116**. This means that delivery tool **110** is configured to deliver the formulation via distal port **115**. In some embodiments, a flexible tube (e.g., flexible tube **134** of FIGS. **1-2**) is configured to connect to connector device **151** so that there is a single fluid path between fluid reservoir **132** of treatment actuator device **130** and distal port **115**. This fluid path can extend from fluid reservoir **132**, through a lumen defined by flexible tube **134**, and through lumen **149** to distal port **115**.

[0109] Since delivery tool **110** is configured to deliver the formulation via the distal opening of the lumen **149** formed by second shaft **116** at distal end **124** and since tip-mounted camera **112** is located at distal end **122** of first shaft **114**, both first shaft **114** and second shaft **116** can simultaneously cross tympanic membrane **30** during a trans-tympanic membrane delivery procedure. This is because when delivery tool **110** delivers a formulation to a targeted site beyond tympanic membrane **30**, a distal portion of first shaft **114** including tip-mounted camera **112** is configured to cross tympanic membrane **30** to provide direct visualization of the targeted site and a distal portion of second shaft **116** is configured to cross tympanic membrane **30** to reach the

targeted site.

[0110] A total length of the delivery tool **110** from a proximal end of the delivery tool **110** to a distal end of delivery tool **110** is equal to length **L4**. In some embodiments, length **L4** may be within a range from 120 mm to 200 mm. In Preferably, length **L4** is equal to approximately 161 mm. When length **L4** is equal to 161 mm or another similar length, this means that delivery tool **110** is sized for clinician to navigate delivery tool **110** to a targeted site within the ear of patient **10** while gripping delivery tool **110** outside of the ear of patient **10**. Handle **118** can have length **L5**. In some embodiments, length **L5** is within a range from 100 mm to 150 mm. Length **L5** of handle **118** can be significantly longer than a length **L1** of first shaft **114** extending distally from distal opening **120** and a length **L2** of a second shaft **116** extending distally from distal opening **120**. In some cases, handle **118** is relatively longer than the exposed portions of first shaft **114** and second shaft **116** because the exposed portions of first shaft **114** and second shaft **116** are sized to fit within the ear of the patient a majority of handle **118** is sized to remain outside of the ear.

[0111] Referring now to FIGS. 7-9, first shaft **114** and second shaft **116** extend distally from distal opening **120** of delivery tool **110**. Delivery tool **110** generally forms a round shape, where a diameter of delivery tool **110** varies along a length of delivery tool **110**. As shown in FIG. 7, for example, a diameter of handle **118** increases from a proximal end of delivery tool **110** to a first peak **145**, decreases to a valley **146** of delivery tool **110**, increases to a second peak **147**, and decreases to a distal end of handle **118**.

[0112] The shape of handle **118** is generally round and extends along a longitudinal axis. In some embodiments, the generally round shape of handle **118** assists clinician **1** in gripping handle **118**. For example, round handles do not include sharp edges that are uncomfortable to grip tightly. Since handle **118** is round, clinician **1** is able to easily rotate a grip so that the distal end of first shaft **114** and the distal end of second shaft **116** are oriented properly within the ear to reach the targeted site.

[0113] In the example of FIGS. 7-9, the shape of handle **118** is generally symmetric about a longitudinal axis of delivery tool **110**. This means that handle **118** does not favor being gripped by a left hand or a right hand. Handle **118** can be gripped equally effectively by a right hand or a left hand. Handle **118** is shaped for a comfortable grip.

[0114] For example, the smooth and shallow region proximate to valley **146** can rest comfortably between the thumb and the index finger of clinician **1**.

[0115] As shown in FIGS. 7-9, first shaft **114** and second shaft **116** are both cylindrical. In some embodiments, a diameter of first shaft **114** is greater than a diameter of second shaft **116**. This is because a distal tip of first shaft **114** can accommodate an image sensor **152** and a light source **154**, the image sensor **152** being wider than the distal port **115** at the distal tip of second shaft **116**. Since the image sensor **152** is placed on a distal end of first shaft **114**, the diameter of first shaft **114** is great enough so that the entire surface area of image sensor **152** fits on the distal end of first shaft **114**. Light source **154** is also located on the distal end of first shaft **114** in the example of FIGS. 7-9.

[0116] In some embodiments, image sensor **152** comprises a camera sensor. Preferably, image sensor **152** comprises a CMOS camera sensor. Image sensor **152** can, in some cases, be rectangular in shape. An array of light-sensitive pixels can be located on a surface of the image sensor **152**. Each of these light-sensitive pixels can record an intensity of light arriving at the pixel. Image data generated by circuitry of the image sensor **152** can include a sequence of image frames, each image frame indicating an intensity value corresponding to each of the pixels. This image data can be displayed on the screen of display device **150** in real time.

[0117] Light source **154** preferably comprises an LED. Optionally, light source **154** includes a waveguide configured to emit light or another kind of light source. The targeted site in the ear of patient **10** for delivering the formulation using delivery tool **110** can be in a dark location that receives little or no natural light and is not visible to the naked eye of an observer. In some embodiments, light source **154** is configured to emit light into the area where the targeted site is

located so that image sensor **152** can capture anatomical features within the area. Image sensor **152** and light source **154** may be part of tip-mounted camera **112**. As shown in FIGS. **7-9**, light source **154** is configured to emit light rays from the distal end of first shaft **114** extending distally beyond the distal end of second shaft **116**. Some of these light rays reflect off objects in and image sensor **152** senses the reflected rays.

[0118] At least part of a portion of second shaft **116** that extends distally beyond a distal end of first shaft **114** is within a field of view of image sensor **152**. This means that image sensor **152** is configured to capture a distal portion of second shaft **116**. When the distal end of first shaft **114** and the distal end of second shaft **116** are both located beyond tympanic membrane **30**, image sensor **152** is therefore configured to capture the location of the distal portion of second shaft **116** relative to a location of one or more anatomical features of the middle ear **40** of patient **10**. This means that the image data captured by image sensor **152** can indicate whether the distal portion of second shaft **116** is positioned to deliver a formulation to a targeted site in the ear of patient **10**.

[0119] Second shaft **116** defines a distal port **115** at a distal end of second shaft **116**. Delivery tool **110** is configured to deliver a formulation to the targeted site through distal port **115** at the distal end of second shaft **116**. The lumen defined by second shaft **116** may extend through an entire length of second shaft **116**. Connector device **151** may connect a flexible tube to the lumen defined by the second shaft **116**. An insertion device may connect to the flexible tube.

[0120] Referring now to FIG. **10**, an example cross-sectional view of first shaft **114** and second shaft **116** shows image sensor **152** and light source **154** located on a tip of first shaft **114** and a cutaway view of second shaft **116**. The view of FIG. **10** is a cutaway view of delivery tool **110** at the distal end of first shaft **114**. This means that the distal end of second shaft **116** including distal port **115** is out of the page relative to the view of FIG. **10**. Image sensor **152** and light source **154** are in the plane of the view of FIG. **10**.

[0121] As shown in FIG. **10**, a surface of first shaft **114** is in contact with a surface of second shaft **116**. A diameter **D1** of first shaft **114** is greater than a diameter **D2** of second shaft **116** in some embodiments. For example, the diameter **D1** of first shaft **114** is 0.5 mm to 1.5 mm, preferably 1.0 mm to 1.4 mm, and 1.3 mm in the depicted embodiment. And in such examples, the diameter **D2** of second shaft **116** is about 0.20 mm to about 0.50 mm, and preferably about 0.41 mm in the depicted embodiment. Also, the diameter **D1** of the first shaft **114** is greater than a maximum lateral width of the image sensor **152** so that the distal-facing end of the first shaft **114** can accommodate the entire surface of image sensor **152**. Image sensor has a first width **W1** and a second width **W2**. In some examples, both of **W1** and **W2** are less than 1.5 mm, and preferably 1 mm or less. In one embodiment, **W1** is equal to 0.575 mm and **W2** is equal to 0.575 mm. Further still, the diameter **D1** of the first shaft **114** is greater than the widths **W1** and **W2** of the image sensor **152** by an amount sufficient to accommodate light source **154**, which is configured to emit light extending distally from a distal end of first shaft **114**. In some examples, first shaft **114** defines a lumen **148** that is sealed at the distal end by a fixed wall (which may comprise an electronics circuit) within the shaft **114** to which image sensor **152** and light source **154** are mounted in an outwardly distal-facing orientation. One or more conductive cables connected to image sensor **152** and light source **154** may extend through lumen **148**.

[0122] Image sensor **152** may be located in a center region of a distal end of first shaft **114** in some embodiments. As shown in FIG. **10**, image sensor **152** is disposed in a center region of the cross-section of first shaft **114**. In some examples, a center point of image sensor **152** is located on a center axis of first shaft **114**. Light source **154**, in some examples, is located in an edge region of a distal end of first shaft **114**. As shown in FIG. **10**, light source **154** is located in an edge region of first shaft **114** above image sensor **152** which is located in a center region of first shaft **114**. An “edge region” of a surface may refer to a region that is displaced from a center of the surface. A “center region” of a surface may refer to a region that includes the center of the surface.

[0123] The formulation delivered by the devices, systems, and methods as described herein can be

a gel, a spray, a mist, a liquid, a paste, a solution, a suspension, an emulsification, and so on, without limitation. In some embodiments, the formulation can contain permeation enhancers or magnetic microparticles to improve the rate of diffusion of the therapeutic agent(s) into the inner ear. In certain embodiments, the formulation can contain lipid encapsulated agents, microparticles, supraparticles or viral vectors, to improve the efficiency and/or extend the duration of the delivery of the therapeutic agent(s) into the inner ear. In particular embodiments, the formulation can contain contrast agents, dyes or stains for diagnostic imaging of the middle and inner ear. In some embodiments, the formulation can comprise or consist of a gel or another material that seals perilymph leakage on the oval window or the round window to treat or prevent a perilymph fistula. [0124] In some embodiments, the formulation or otic composition (e.g., an extended-release otic composition) can be delivered to a subject from or with the help of the treatment devices described herein. Such a formulation may be delivered using an implantable formulation carrier such as an implant device, or by directly injecting or otherwise delivering the formulation.

[0125] In some embodiments, an extended-release formulation can include a polymer composition that can form a gel. For example, a polymer composition can include a functional polymer, wherein the functional polymer includes a first functional group, and a crosslinker, wherein the crosslinker includes a second functional group, and water, wherein a crosslinking reaction can occur between the first functional group and the second functional group to form a gel. In some embodiments, the functional polymer can be present in an amount of about 5% to about 15% by weight of the polymer composition. In some embodiments, the crosslinker can be present in an amount of about 0.2% to about 0.6% by weight of the polymer composition.

[0126] It will be appreciated that a first functional group (e.g., on a functional polymer) and a second functional group (e.g., on a crosslinker) should be such that a crosslinking reaction can occur. Therefore, the choice of functional polymer can be based on the choice of crosslinker, or vice versa. In some embodiments, a first functional group can be an N-hydroxysuccinimide (NHS) group and a second functional group can be an amine (e.g., a primary amine), or vice versa. In some cases, the functional polymer contains only electrophilic or nucleophilic functional groups, and the crosslinker contains only nucleophilic or electrophilic functional groups, respectively.

[0127] In some embodiments, the functional polymer is a multi-arm (e.g., 3-arm, 4-arm, 6-arm, or 8-arm) polyethylene glycol (PEG) including two or more succinimidyl ester (e.g., a succinimidyl succinate or a succinimidyl glutarate) or sulfo-succinimidyl ester functional groups and the crosslinker contains a plurality of amine (e.g., primary amine) functional groups. In some embodiments, the multi-arm PEG can have two or more arms terminate in a succinimidyl ester functional group. In some embodiments, one or more monomers of the multi-arm PEG can include a succinimidyl ester functional group. In some embodiments, the crosslinker can be a polylysine (e.g., an epsilon-polylysine) (e.g., trilysine, tetralysine, or pentalysine). For example, in some embodiments, the functional polymer can be pentaerythritol poly(ethylene glycol) ether tetrasuccinimidyl glutarate, and the crosslinker can be trilysine.

[0128] In some embodiments, the functional polymer is a multi-arm (e.g., 3-arm 4-arm, 6-arm, or 8-arm) polyethylene glycol including two or more amine (e.g., primary amine) functional groups and the crosslinker includes a plurality of succinimidyl ester (e.g., a succinimidyl succinate or succinimidyl glutarate) or sulfo-succinimidyl ester functional groups. In some embodiments, the multi-arm PEG can have two or more arms terminate in an amine (e.g., primary amine) functional group. In some embodiments, one or more monomers of the multi-arm PEG can include an amine (e.g., primary amine) functional group. In some embodiments, the crosslinker can be disuccinimidyl glutarate, disuccinimidyl suberate, bis(sulfosuccinimidyl) suberate, or disuccinimidyl succinate.

[0129] In some embodiments, an extended-release otic composition can include an active agent (e.g., a therapeutic agent, a prophylactic agent, a diagnostic or visualization agent, or a combination thereof). An active agent can include, for example, a protein (e.g., an enzyme, a growth factor, an

antibody or an antigen-binding fragment thereof), a carbohydrate (e.g., a glycosaminoglycan), a nucleic acid (e.g., an antisense oligonucleotide, an aptamer, a micro RNA, a short interfering RNA, or a ribozyme), small molecules, or combinations thereof. In some embodiments, a small molecule can include an antibiotic, an antineoplastic agent (e.g., doxorubicin), a local anesthetic, a steroid, a hormone, an apoptotic inhibitor, an angiogenic agent, an anti-angiogenic agent (e.g., a VEGF inhibitor), a neurotransmitter, a neuroprotectant, a psychoactive drug, an anti-inflammatory, and combinations thereof.

[0130] In some embodiments, an active agent of the formulation can include an anti-angiogenic agent. In some embodiments, an anti-angiogenic agent can be a VEGF inhibitor. In some cases, a VEGF inhibitor can be an antibody or an antigen-binding fragment thereof, a decoy receptor, a VEGFR kinase inhibitor, an allosteric modulator of a VEGFR, or a combination thereof. In some cases, a VEGF inhibitor can be an antibody or an antigen-binding fragment thereof. For example, in some embodiments, a VEGF inhibitor can be alacizumab, bevacizumab (AVASTIN®), icrucumab (IMC-18F1), ramucirumab (LY3009806, IMC-1121B, CYRAMZA®), or ranibizumab (LUCENTIS®). In some embodiments, a VEGF inhibitor can be a decoy receptor (e.g., aflibercept). In some embodiments, a VEGF inhibitor can be a VEGFR kinase inhibitor, such as agerafenib, altiratinib, apatinib, axitinib, cabozantinib, cediranib, lapatinib, lenvatinib, motesanib, nintedanib, pazopanib, pegaptanib, rebastinib, regorafenib, semaxanib, sorafenib, sunitinib, toceranib, tivozanib, or vandetanib. Other examples of VEGF inhibitors may be known in the art. In some embodiments, a VEGFR inhibitor can be an allosteric modulator of a VEGFR (e.g., cyclotraxin B).

[0131] An extended release formulation or otic composition can, in some cases, be useful to treat an otic disease or disorder, such as Ménière's Disease (MD), Autoimmune Inner Ear Disease (AIED), sudden sensorineural hearing loss (SSNHL), noise-induced hearing loss (NIHL), age-related hearing loss, sensorineural hearing loss associated with diabetes, tinnitus, damaged cilia from an autoimmune disorder, damaged cilia from an infection, damaged cilia from excess fluid or pressure, hearing loss due to chemotherapy, or a combination thereof.

[0132] Formulations that can be delivered from or with the help of the treatment devices described herein can also include but are not limited to antioxidants, anti-inflammatory steroids, antimicrobials, NMDA receptor antagonists, nootropics, anti-apoptotic agents, neurotrophins, neuroprotective agents, neural protective proteins such as CNTF, BDNF, PEDF, NGF, NT-3, and the like, cannabinoids, monoclonal antibodies, other proteins, gene therapy, iRNA, tyrosine kinase inhibitors (TKIs), dual leucine zipper kinase (DLK) inhibitors, and protein therapies like anti-VEGF.

[0133] As an example, the therapeutic agent of the formulation can include, but is not limited to antimicrobials such as antibiotics such as tetracycline, chlortetracycline, bacitracin, neomycin, polymyxin, gramicidin, cephalixin, oxytetracycline, chloramphenicol kanamycin, rifampicin, ciprofloxacin, tobramycin, gentamycin, erythromycin and penicillin; antifungals such as amphotericin B, clotrimazole, ketoconazole and miconazole; anti-bacterials such as sulfonamides, sulfadiazine, sulfacetamide, sulfamethizole and sulfisoxazole, nitrofurazone and sodium propionate; antivirals such as idoxuridine, trifluorotymidine, acyclovir, ganciclovir and interferon; antiallergenics such as sodium cromoglycate, antazoline, methapyrilene, chlorpheniramine, pyrilamine, cetirizine and propenpyridamine; anti-inflammatories such as hydrocortisone, hydrocortisone acetate, dexamethasone, dexamethasone 21-phosphate, fluocinolone, medrysone, prednisolone, prednisolone 21-phosphate, prednisolone acetate, fluoromethalone, betamethasone, and triamcinolone; non-steroidal anti-inflammatories such as salicylate, indomethacin, ibuprofen, diclofenac, flurbiprofen and piroxicam; decongestants such as phenylephrine, naphazoline and tetrahydrozoline; miotics and anticholinesterases such as pilocarpine, salicylate, acetylcholine chloride, physostigmine, eserine, carbachol, diisopropyl fluorophosphate, phospholine iodide and demecarium bromide; mydriatics such as atropine sulfate, cyclopentolate, homatropine,

scopolamine, tropicamide, eucatropine and hydroxyamphetamine; sympathomimetics such as epinephrine; antineoplastics such as carmustine, cisplatin and fluorouracil; immunological drugs such as vaccines and immune stimulants; hormonal agents such as estrogens, estradiol, progestational, progesterone, insulin, calcitonin, parathyroid hormone and peptide and vasopressin hypothalamus releasing factor; beta adrenergic blockers such as timolol maleate, levobunolol HCl and betaxolol HCl; growth factors such as epidermal growth factor, fibroblast growth factor, platelet derived growth factor, transforming growth factor beta, somatotropin and fibronectin; carbonic anhydrase inhibitors such as dichlorophenamide, acetazolamide and methazolamide and other drugs such as prostaglandins, antiprostaglandins and prostaglandin precursors; keratolytic agents such as selenium sulfide, imiquimod, salicylic acid, and retinoids; antioxidants, NMDA receptor antagonists, nootropics, anti-apoptotic agents, neurotrophins, neuroprotective agents, tyrosine kinase inhibitors (TKIs), dual leucine zipper kinase (DLK) inhibitors, Rho-kinase (ROCK) inhibitors, cannabinoids, monoclonal antibodies, antibody fragments, other proteins, and gene therapy. Other therapeutic agents known to those skilled in the art which are capable of controlled, sustained release into the ear in the manner described herein are also suitable for use in accordance with embodiments of the devices described herein.

[0134] The therapeutic agent of the formulation can include, but is not limited to sodium thiosulfate to protect against cisplatin-induced hearing loss; thiouracil to protect against cisplatin-induced hearing loss; NMDA receptor antagonists for the treatment of tinnitus (AM-101; Auris Medical); AM-111 containing the synthetic peptide D-JNKI-1 (D-stereoisomer of c-Jun N-terminal Kinase Inhibitor 1; Auris Medical) for otoprotection in acute inner ear hearing loss; dexamethasone and other corticosteroids for the treatment of Meniere's Disease and forms of vestibular disorders and/or hearing loss associated with inflammation; D-methionine (Southern Illinois University) to protect against Noise-induced hearing loss; PIPE-505, LY411575, and LY3056480 (selective gamma secretase inhibitors that block Notch activation); Verosudil (AR-12286) and Netarsudil to treat synaptopathy; BDNF to treat synaptopathy; NGF; and NT-3 neurotrophic factor.

[0135] The therapeutic agent of the formulation can include but is not limited to local anesthetics for delivery into the ear canal including benzocaine, antipyrine, butamben, dibucaine, lidocaine, prilocaine, oxybuprocaine, pramoxine, proparacaine, proxymetacaine, and tetracaine.

[0136] Various pharmaceutically acceptable carriers for the therapeutic agents described herein can include such as, for example, solids such as starch, gelatin, sugars, natural gums such as acacia, sodium alginate and carboxymethyl cellulose; polymers such as silicone rubber; liquids such as sterile water, saline, dextrose, dextrose in water or saline; condensation products of castor oil and ethylene oxide, liquid glyceryl triester of a lower molecular weight fatty acid; lower alkanols; oils such as corn oil, peanut oil, sesame oil, castor oil, and the like, with emulsifiers such as mono- or di-glyceride of a fatty acid, or a phosphatide such as lecithin, polysorbate **80**, and the like; glycols and polyalkylene glycols including **P407** and other combinations of polyethylene glycol and polypropylene glycol; aqueous media in the presence of a suspending agent, for example, sodium carboxymethylcellulose, hyaluronic acid, sodium hyaluronate, sodium alginate, poly(vinyl pyrrolidone) and similar compounds, either alone, or with suitable dispensing agents such as lecithin, cyclodextrins, polyoxyethylene stearate and the like. The carrier may also contain adjuvants such as preserving, stabilizing, wetting, emulsifying agents or other related materials.

[0137] A therapeutic agent referred to with a trade name encompasses one or more of the formulation of the therapeutic agent commercially available under the tradename, the active ingredient of the commercially available formulation, the generic name of the active ingredient, or the molecule comprising the active ingredient. As used herein, a therapeutic or therapeutic agents are agents that ameliorate the symptoms of a disease or disorder or ameliorate the disease or disorder. Therapeutic agent, therapeutic compound, therapeutic regimen, or chemotherapeutic include conventional drugs and drug therapies, including vaccines, which are known to those skilled in the art and described elsewhere herein. Therapeutic agents include, but are not limited to,

moieties that are capable of controlled, sustained release into the body.

[0138] While the devices, systems, materials, compounds, compositions, articles, and methods described herein described in the context of treating hearing loss, it should be understood that the devices, systems, materials, compounds, compositions, articles, and methods may be used to treat any disorder of the middle ear and/or inner ear including, but not limited to, tinnitus, balance disorders including vertigo, Meniere's Disease, vestibular neuronitis, vestibular schwannoma, labyrinthitis, otosclerosis, ossicular chain dislocation, cholesteatoma, otitis media, middle ear infections, and tympanic membrane perforations, to provide a few examples.

[0139] Although the round window membrane is one targeted site for therapeutic agent delivery or access, the systems and methods described herein can also be used for precise delivery of therapeutic agents to other targeted sites, such as the oval window or other parts of the middle ear cavity, and for providing access to other features or regions of the middle ear. For example, the systems and methods described herein can be used for minimally invasive surgical reconstruction of the ossicular chain, for removal of cholesteatoma, for diagnostic assessment, and other procedures. Any and all such techniques for using the systems and methods described herein are included within the scope of this disclosure.

[0140] The devices, systems, materials, compounds, compositions, articles, and methods described herein may be understood by reference to the above detailed description of specific aspects of the disclosed subject matter. It is to be understood, however, that the aspects described above are not limited to specific devices, systems, methods, or specific agents, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

[0141] A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the claim scope here. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A system for delivering a therapeutic formulation at a targeted site in an ear of a patient, wherein the system includes an otologic delivery tool comprising: a handle connected to side-by-side trans-tympanic shafts that are fixedly mounted in a stationary position relative to one another and extend distally from the handle, wherein the side-by-side trans-tympanic shafts comprise a visualization shaft and a therapeutic formulation delivery shaft having a distal port positioned distally of the visualization shaft; a tip-mounted image capture device fixedly mounted at a distal end of the visualization shaft and being connected with an image data cable extending through a lumen of the visualization shaft proximally toward the handle; and a fluid input port on the handle to connect with a treatment actuator device to provide fluid communication to a fluid delivery lumen of the therapeutic formulation delivery shaft, wherein the distal port of the therapeutic formulation delivery shaft is configured to deposit the therapeutic formulation at the targeted site in the ear of the patient while the tip-mounted image capture device provides direct visualization of the therapeutic formulation delivery shaft.
2. The system of claim 1, wherein the visualization shaft and the therapeutic formulation delivery shaft are fixedly mounted in a stationary position relative to one another in a parallel relationship.
3. The system of claim 1, wherein the visualization shaft is in side-by-side contact with the therapeutic formulation delivery shaft along an entire length of the visualization shaft extending distally from the handle.
4. The system of claim 1, wherein the visualization shaft and the therapeutic formulation delivery shaft are both fixedly mounted to a distal end of the handle.
5. The system of claim 4, wherein the visualization shaft is fixedly mounted to the distal end of the handle and extends distally from the distal end of the handle for a first distal length, wherein the

therapeutic formulation delivery shaft is fixedly mounted to the distal end of the handle and extends distally from the distal end of the handle for a second distal length that is greater than the first distal length such that the tip-mounted image capture device distal end of the visualization shaft is positioned in a distal-facing orientation to provide direct visualization of a distal tip of the therapeutic formulation delivery shaft.

6. The system of claim 5, wherein the second distal length that is greater than the first distal length such that a distal tip of the therapeutic formulation delivery shaft extends beyond the visualization shaft and the tip-mounted image capture device at the distal end of the visualization shaft is positioned in a distal-facing orientation to provide direct visualization of the distal tip of the therapeutic formulation delivery shaft.

7. The system of claim 1, further comprising the treatment actuator device, wherein the treatment actuator device includes a fluid reservoir containing the therapeutic formulation.

8. The system of claim 7, further comprising a flexible tube extending from the treatment actuator device to the fluid input port of the handle, wherein a connector device is configured to connect a lumen of the flexible tube with the lumen of the delivery shaft.

9. The system of claim 7, wherein the treatment actuator device is configured to direct connect to the fluid input port of the handle.

10. The system of claim 1, further comprising a video controller configured to receive image data from the tip-mounted image capture device and including at least one video output port to connect with a user interface display to present a location of a distal tip of a therapeutic formulation delivery shaft relative to the targeted site in the ear.

11. The system of claim 10, wherein the tip-mounted image capture device comprises: a complementary metal-oxide semiconductor (CMOS) image sensor having a maximum lateral width of less than 1 mm, a light-emitting diode (LED) configured to output an emitted optical signal, wherein the CMOS image sensor is configured to detect a reflected optical signal, and circuitry configured to generate the image data based on the emitted optical signal and the reflected optical signal.

12. The system of claim 11, wherein a combined maximum lateral width of the CMOS image sensor and the LED extends parallel to an outer diameter of the visualization shaft and is less than the outer diameter of the visualization shaft.

13. The system of claim 11, wherein the CMOS image sensor is located in a center of the distal end of the visualization shaft and is fixed in a distal-facing orientation, and wherein the LED is located laterally offset from the CMOS image sensor at the distal end of the visualization shaft and is fixed in a distal-facing orientation.

14. The system of claim 1, wherein a distal tip of the therapeutic formulation delivery shaft extends distally beyond the visualization shaft distal portion by a distal extension distance of 1 mm to 5 mm.

15. The system of claim 14, wherein the visualization shaft is fixedly mounted to the distal end of the handle and extends distally from the distal end of the handle for a first distal length of 20 mm to 30 mm, the therapeutic formulation delivery shaft is fixedly mounted to the distal end of the handle and extends distally from the distal end of the handle for a second distal length of 22.5 mm to 33.5 mm, and the handle has an axial length of 100 mm to 150 mm that extends parallel to the visualization shaft and the therapeutic formulation delivery shaft.

16. A system for delivering a formulation at a targeted site in an ear of a patient, wherein the system includes: a treatment actuator device; a formulation delivery tool comprising: an image capture device; a visualization shaft comprising a visualization shaft distal portion, wherein the image capture device is located at the visualization shaft distal portion; a delivery shaft defining a lumen, the delivery shaft comprising a delivery shaft distal portion extending distally beyond the visualization shaft distal portion, wherein the visualization shaft and the delivery shaft are aligned so that a location of the delivery shaft distal portion is stationary relative to a location of the

visualization shaft distal portion, and wherein the delivery shaft is configured to deliver, via the lumen from the treatment actuator device, the formulation at the targeted site in the ear of the patient as the image capture device captures the delivery shaft distal portion; and a flexible tube connecting the treatment actuator device to the lumen of the delivery shaft.

17. The system of claim 16, wherein the visualization shaft and the delivery shaft are parallel.

18. The system of claim 16, wherein the image capture device comprises: an image sensor; and a light-emitting diode (LED) configured to output an emitted optical signal, wherein the image sensor is configured to detect a reflected optical signal; and circuitry configured to generate image data based on the emitted optical signal and the reflected optical signal.

19. The system of claim 18, wherein the image sensor is located in a center region of a distal end of the visualization shaft, and wherein the LED is located in an edge region of the distal end of the visualization shaft, the edge region displaced from the center region.

20. A system for delivering a formulation to a round window niche of a patient, wherein the system includes: a treatment actuator device; a formulation delivery tool comprising: an image capture device; a visualization shaft comprising a visualization shaft distal portion, wherein the image capture device is located at the visualization shaft distal portion; and a delivery shaft defining a lumen, the delivery shaft comprising a delivery shaft distal portion extending distally beyond the visualization shaft distal portion, and wherein the delivery shaft is configured to deliver, via the lumen from the treatment actuator device, the formulation at the round window niche of the patient as the image capture device captures the delivery shaft distal portion; and a flexible tube connecting the treatment actuator device to the lumen of the delivery shaft.
